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Invidia-smi
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Tue May 5 05:28:03 2026
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|                            |          |      |                           |                  |              |                    |            |     |
|----------------------------|----------|------|---------------------------|------------------|--------------|--------------------|------------|-----|
| NVIDIA-SMI 580.82.07       |          |      | Driver Version: 580.82.07 |                  |              | CUDA Version: 13.0 |            |     |
| GPU                        | Name     |      | Persistence-M             | Bus-Id           | Disp.A       | Volatile           | Uncorr.    | ECC |
| Fan                        | Temp     | Perf | Pwr:Usage/Cap             |                  | Memory-Usage | GPU-Util           | Compute M. |     |
|                            |          |      |                           |                  |              |                    | MIG M.     |     |
| 0                          | Tesla T4 |      | Off                       | 00000000:00:04.0 | Off          |                    | 0          |     |
| N/A                        | 33C      | P8   | 9W / 70W                  | 0MiB / 15360MiB  |              | 0%                 | Default    | N/A |
|                            |          |      |                           |                  |              |                    |            |     |
| Processes:                 |          |      |                           |                  |              |                    |            |     |
| GPU                        | GI       | CI   | PID                       | Type             | Process name | GPU Memory         |            |     |
|                            | ID       | ID   |                           |                  |              | Usage              |            |     |
| No running processes found |          |      |                           |                  |              |                    |            |     |

```
# =====
# 1. IMPORT LIBRARIES
# =====
import tensorflow as tf
from tensorflow.keras import layers, models
from tensorflow.keras.preprocessing.sequence import pad_sequences

# =====
# 2. LOAD DATASET (IMDB)
# =====
vocab_size = 10000

(x_train, y_train), (x_test, y_test) = tf.keras.datasets.imdb.load_data(num_words=vocab_size)

print("Training samples:", len(x_train))
print("Testing samples:", len(x_test))

# =====
# 3. PREPROCESSING
# =====
max_len = 200

x_train = pad_sequences(x_train, maxlen=max_len, padding='post')
x_test = pad_sequences(x_test, maxlen=max_len, padding='post')

# =====
# 4. BUILD RNN MODEL (FIXED)
# =====
model = models.Sequential([

    # Input layer (fixes unbuilt issue)
    layers.Input(shape=(max_len,)),

    # Embedding layer
    layers.Embedding(input_dim=vocab_size, output_dim=128),

    # Simple RNN layer
    layers.SimpleRNN(64),

    # Dense layer
    layers.Dense(64, activation='relu'),

    # Dropout
    layers.Dropout(0.5),

    # Output layer
    layers.Dense(1, activation='sigmoid')
])

# Now summary will show correct parameters
model.summary()

# =====
# 5. COMPILE MODEL
# =====
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model.compile(
    optimizer='adam',
    loss='binary_crossentropy',
    metrics=['accuracy']
)

# =====
# 6. TRAIN MODEL
# =====
history = model.fit(
    x_train, y_train,
    epochs=38,
    batch_size=32,
    validation_split=0.2
)

# =====
# 7. EVALUATE MODEL
# =====
loss, accuracy = model.evaluate(x_test, y_test)

print("\nTest Accuracy:", accuracy * 100)

# =====
# 8. PREDICT NEW TEXT
# =====
word_index = tf.keras.datasets.imdb.get_word_index()

def encode_text(text):
    words = text.lower().split()
    encoded = [word_index.get(word, 2) for word in words]
    padded = pad_sequences([encoded], maxlen=max_len, padding='post')
    return padded

def predict_sentiment(text):
    processed = encode_text(text)
    prediction = model.predict(processed)[0][0]

    print("\nReview:", text)

    if prediction >= 0.5:
        print("Sentiment: Positive 😊")
    else:
        print("Sentiment: Negative 😞")

# Example predictions
predict_sentiment("This movie was excellent and amazing")
predict_sentiment("This movie was very bad and boring")
```

